

# GMRCK25S Rear Helper Spring Kit Installation Manual

## Model: GMRCK25S (Rear Suspension Package)

Please read these instructions carefully before starting the installation. Ensure all standard safety precautions are observed while working underneath the vehicle.

### 1. Parts List

Before beginning, verify that all components listed below are present in your kit.

| Ref # | Description / Specifications                       | Qty     |
|-------|----------------------------------------------------|---------|
| 1     | Hex Cap Screw (HCS) 3/8" x 1-1/4" UNF Grade 5 Zinc | 4       |
| 2     | Weldment Bracket (Left-Hand & Right-Hand)          | 1L - 1R |
| 3     | Flat Washer USS 3/8" x 1" Grade 2 Zinc             | 4       |
| 4     | Nylock Locknut 3/8" UNF Grade 5 Zinc               | 4       |
| 5     | Rubber Spacer (4" O.D. x 1" Thick with AH Bushing) | 2       |
| 6     | Hollow Rubber Spring                               | 2       |
| 7     | Hex Cap Screw (HCS) 3/8" x 2-3/4" UNC Grade 5 Zinc | 2       |
| 8     | Hex Cap Screw (HCS) 3/8" x 1-3/4" UNC Grade 5 Zinc | 2       |

### 2. Installation Instructions

- 1. Raise Vehicle:** Raise the vehicle using an appropriate hoist or jack, allowing the axle to hang freely. Support the chassis securely before proceeding.
- 2. Remove Bump Stop:** Remove the existing OEM vehicle bump stop and its associated bracket.
- 3. Assemble Helper Spring Unit:** Assemble the hollow rubber spring and the weldment bracket. Refer to any provided diagrams for correct orientation.
- 4. Mount Assembly:** Install the complete helper spring assembly onto the vehicle frame utilizing the existing mounting holes and the provided fasteners.
- 5. Lower Vehicle:** Carefully lower the vehicle back to the ground.

#### Installation Notes:

- The included 1" rubber spacer (Ref 5) is optional. Use it only when needed to reduce excessive unloaded clearance. Use the longer 3/8" x 2-3/4" hex cap screw (Ref 7) with the spacer and the shorter 3/8" x 1-3/4" hex cap screw (Ref 8) without it.
- Verify that vehicle brake lines or cables do not interfere with the helper spring's position or movement.
- Ensure the weldment bracket (Ref 2) is oriented with the larger disc side facing towards the vehicle's tire.

### 3. Clearance Instructions & Fitting Guidance

- Clearance Gap:** With the vehicle unloaded and at normal ride height, maintain approximately **1/2" to 1-1/2" (13-38 mm)** of clearance between the helper spring and the vehicle contact point. Approximately **1" (25 mm)** is a practical target.
- Ride Comfort:** Proper clearance helps maintain a comfortable unloaded ride. Insufficient clearance or excessive preload may result in a firmer ride.
- Spacer Use:** If the unloaded gap is too large, install the included **1" spacer** to reduce it. If the clearance is already within the recommended range, the spacer is not required. If the spacer causes little or no clearance, remove it.
- Load Effects:** A partial load, cargo, or aftermarket vehicle accessories may reduce or eliminate this initial clearance.

**Important Safety Warnings:**

- Do not exceed the vehicle manufacturer's recommended axle weight rating / GAWR.
- Inspect for potential interference between the hollow rubber spring and adjacent vehicle components. Under load, the rubber spring will expand in diameter.

*Note: Illustrations are for reference only. Vehicle configuration and actual component appearance may vary. Install according to the vehicle-specific mounting points and standard mechanical practices.*

**INSTALLATION DIAGRAM**